

Fine-tuning strategy for a verified TLA+ prover: measured position and options

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Prepared 2026-08-12 for the fine-tuning strategy discussion. All numbers are from ledger re-scores of `rows.jsonl` on the frozen 30-spec holdout unless stated otherwise. A composition arm is training on Argonne as this is written; its result is not included and is not predicted here.

Summary

Six supervised fine-tunes and seven RL runs have been measured. One arm beats the untuned base: W4-diamond-gold, at 13.5% per-sample against 6.9% (+0.086, $p=0.063$, paired bootstrap on byte-identical prompts). Every RL run failed for the same mechanical reason, and that reason is the same fact that limits the SFT arms: between 68% and 74% of failed samples do not parse, against 2.1% that model the system incorrectly. The corpus contains no instance of that failure mode.

Corrections to the agenda's stated position

Four points in the supplementary material are out of date or not supported by the primary evidence. Raising them first because they change what the meeting should decide.

chattla-v2-sft2 is not the most recent or the best model. It is the arm that failed Gate-2: framing A flat at 11/30 against base 12/30, framing B a real regression (pass@1 6/23 against baseline 15/23). Three arms have been trained since. The best measured checkpoint is **chattla-w4dg-120b** (W4-diamond-gold), which is the only fine-tune to beat base on per-sample yield.

The published 30% Gold / 9-of-30 result has three defects. Verified directly against the repo and logs: (1) all 30 holdout module names appear in the GRPO topic pool `diamond_gen_topics.json`, and the RL dataset loader implements no holdout filter, so the models received TLC-verifier reward on the descriptions of the exact 30 evaluation tasks; the SFT path does filter, so this is specific to the RL loaders; (2) the eval is a four-shot repair loop with verifier feedback, and cross-tabulating result against `attempts_used` gives a single-shot rate of 3/30; (3) the LoRA merge failed with an argument error five seconds before the eval that names the merged Ollama tag, so the evaluated checkpoint cannot be confirmed as the GRPO-trained weights. This number should not be carried into new work without those three qualifiers.

“RL into higher pass rates” has already been attempted seven times. All failed, and the failure is diagnosed rather than suspected. Detail in the RL section below. The short version: GRPO needs reward variance within a sampled group, and when nearly every completion fails to parse, every completion in the group scores identically and the gradient is zero. RL is blocked behind parse competence, not behind reward design; three reward designs were tried and all three starved.

tla-w4-diamond-gold does contain commented code, though sparsely: 38.7% of targets contain at least one comment, with a median of 7 comment lines in those targets, and an overall comment-line density of 3.9% against a median target length of 75 lines. The concern behind the claim is reasonable — comment density is low, and there is no supervision teaching the model to explain its own specs — but “lacks any commented code” overstates it.

The vacuity and reward-hacking concern is better founded and quantified below.

Measured arms

Framing A, k=32, 30-spec frozen holdout. Per-sample rate is the primary statistic; pass@32 is reported second because it discards roughly 97% of the evidence and has twice produced misleading conclusions in this program.

arm	corpus	per-sample	pass@32	vs base
untuned gpt-oss-120b	—	6.9% (66/960)	12/30	—
v2 SFT	260 harmony pairs	4.9% (47/963)	11/30	worse
W4-diamond-gold	3,534 rows, bare rendering	13.5% (130/960)	16/30	+0.086, p=0.063
W4DG-genprompt	4,219 rows, prompt-aligned	7.7% (69/896)	12/30	−0.018, p=0.66
mech (composition)	6,906 rows incl. repair	training	—	pending

Framing B (repair from a corrupted spec) per-sample: untuned baseline 55.0%, v2 SFT 28.7%, W4DG-genprompt 20.5%. Every fine-tune so far has *degraded* repair performance relative to the untuned model, which is the strongest single argument that the corpus composition, not the volume, is what needs to change.

External reference points from the same audit: Opus 4.7 zero-shot reaches 7/30 (23.3%) non-vacuous on the FormaLLM validation set, and the best untuned model in a 30-model sweep reached 8.6% semantic pass on 205 specs. Frontier prompting is currently ahead of every fine-tune we have produced.

Two negative results

Prompt alignment. W4DG-genprompt tested the hypothesis that training pairs should use the same prompt the specs were verified under. It regressed to base (13.5% → 7.7%) and raised **sany=fail** from 68.1% to 74.1%. The candidate mechanism, untested, is that training on one rigid template binds the capability to that template while Gate-2 uses a third prompt shape. The practical lesson is that rendering choices are not cosmetic and can erase a real gain.

Parse recovery is not pass recovery. A deterministic lint pass (drop builtin-colliding declarations, add missing EXTENDS, dedupe definitions) made 88 of 454 failed candidates parse across 9 unsolved specs, and flipped zero specs to passing. Making output parse is necessary and demonstrably not sufficient.

The RL era, and why it stopped

Seven runs on gpt-oss-20b, 2026-03 to 2026-04, on two 49GB RTX 8000s.

run	method	outcome
per-action	GRPO	no surviving artifacts; flat reward reported second-hand
piecewise	DPO	final loss 0.6661 against ln 2 = 0.6931, accuracy 0.60
full-spec	GRPO	172 steps, mean reward 0.0511, 37 of 172 steps nonzero
repair R1	GRPO	965 steps, mean reward 0.1487, 703 steps at zero variance
repair R2	GRPO	600 steps, mean 0.0930, 430 at zero variance
repair R3	—	aborted pre-training: 152 usable pairs against a floor of 300
repair (July)	GRPO	89 steps at uniform zero reward; after a fix, 1 of 7 rows improved

frac_reward_zero_std equals 1 on the majority of steps in every GRPO run. Repair R1’s mean reward of 0.1487 is the shaping function’s hard-coded “no change” constant of 0.15, so the run sat on the flat part of its own reward for 965 steps. KL stayed between 0.002 and 0.014, so this was not policy collapse under KL. There is no evidence of reward hacking and none of verifier false positives.

The consequence for strategy: **reward variance is a function of parse rate**. RL becomes viable at the point where a meaningful fraction of a sampled group parses, and not before. Sequencing RL after mechanics competence is not a preference, it is a precondition.

What “100%” can mean

The program’s locked north-star is guaranteed-correct TLA+ obtained through the verification loop, not through the weights. That distinction should be explicit in the meeting, because it determines which metric the fine-tuning strategy optimizes.

A model that emits correct TLA+ 100% of the time on the first try is not a realistic target and is not needed. What the loop needs is a low expected number of iterations to a verified spec. Framed that way:

- 100% correctness is supplied by the verifier, which is sound and already in place.
- The weights’ job is to reduce iterations and to keep the loop from stalling. A spec that never parses cannot be repaired productively, which is the observed stall mode.
- The right headline metric is therefore expected-iterations-to-verified, or equivalently the pass@k curve’s shape, not single-shot pass rate. We do not currently report this, and adopting it would change which arms look good. It is worth deciding at the meeting whether to adopt it.

Dataset assessment: tla-w4-diamond-gold

Composition of the 4,119-row cut: `mutex_locks` 35.5%, other 18.0%, consensus 15.8%, commit protocols 8.9%, queues and buffers 8.9%. Liveness rows are 12.4% of the render against a public TLA+ liveness rate of 23–34%.

The vacuity and reward-hacking concern raised in the agenda is supported: the real mutation-catch rate is 12.7%, meaning most specs in the corpus do not have invariants strong enough to detect a single-operator mutation. A spec that passes TLC with a weak invariant is a verified spec that teaches little. The mutation-gated “diamond” tier exists to address this and is the tier that measured best in the TLA-Prover ablation, which is consistent with the concern being real.

Scenario diversity, by contrast, was probably never the bottleneck: W4-diamond-gold varied noun, scenario and number combinatorics heavily, and the failure profile barely moved. The evidence points at what the corpus *demonstrates* (only already-correct specs) rather than at how many distinct scenarios it covers.

Options, with predicted effect

Ordered by evidence strength, not by appeal. Each is a single knob; the program’s worst results came from turning several at once.

1. Mechanics supervision — in flight. Add verified broken-to-fixed pairs so the corpus contains the dominant failure mode. 2,787 pairs were manufactured from the existing survivor corpus by corrupting a spec, confirming with SANY and TLC that the corruption really fails, and using the original text as the repair target. No teacher model, no rejection sampling, minimal-by-construction fixes. Result expected within a day. This is the direct attack on the 68–74%.

2. Grammar-constrained decoding — untested, no training cost. If parse failure is mechanical, constraining generation to the TLA+ grammar attacks it without a training run at all. Two EBNF grammars are already scaffolded in the harness (`tla_module_v0.ebnf`, `tla_proof_v0.ebnf`). This is attractive precisely because it is orthogonal to the training arms and cannot confound them. It is the cheapest untried lever in the program.

3. Multi-task environment split — untested. The strongest positive result in the Meta FAIR synthetic-data work is that splitting a fixed budget across four task environments beats concentrating it on one, on both pass@10 and out-of-distribution generalization. We train one shape at a time and already own verifiers for four: generation, repair, abduction (counterexample trace to violated invariant), and fuzzing (the mutation battery inverted). Amendment 16 treated task-shape mismatch as a confound to eliminate; this evidence says it may be a lever to use.

4. RL, after (1) or (2) lands. Improvement-reward GRPO on repair is the natural fit, and the existing implementation is reusable. It should not be restarted until a sampled group produces score variance; the honest test is to measure `frac_reward_zero_std` on a candidate checkpoint before committing compute.

5. More data of the current kind — weakest. Going from 260 to 4,119 rows roughly doubled per-sample yield but left the failure profile unchanged. Volume has visible diminishing returns against a corpus that omits the failure mode.

Method and infrastructure notes

Eleven defects have been found and fixed across the training and measurement paths; four of the six training defects exited 0 on runs that had not trained what they were supposed to train. The two that matter for anyone reading old results: a gpt-oss-specific LoRA selector left a dense model's FFN frozen at 0.0195% trainable while reporting success, and a prompt-construction bug under-specified 13 of 30 framing-A prompts, corrupting every framing-A measurement recorded before 2026-07-29. Numbers from before those fixes should not be compared against numbers from after them.

Current guards: an aborting trainable-parameter floor, attach-time expert-coverage reporting, exit-code propagation through PBS, a mandatory preflight against real weights, ledger-only scoring, and derived-seed de-code provenance.

Questions for the meeting

1. Do we adopt expected-iterations-to-verified as the headline metric, in place of single-shot pass rate?
2. Is grammar-constrained decoding worth doing before more fine-tuning, given it needs no training compute and cannot confound the training arms?
3. Should the next corpus investment go to task-shape diversity (four environments) or to depth in generation and repair alone?
4. The 30-spec holdout is the binding constraint on statistical power: effects below roughly 4 points cannot be resolved. Widening it requires an amendment and a decontamination pass. Do we spend that?
5. What is published, and with which qualifiers, given the three defects under the existing 30% result?